dies, it stops absorbing CO_2. Since ${}^{14}_{6}C$ starts to decay, the ratio of ${}^{14}_{6}C$ to ${}^{12}_{6}C$ in a dead organism or specimen decreases over the years. Suppose the ratio of ${}^{14}_{6}C$ to ${}^{12}_{6}C$ in the ancient tree pieces excavated is known, then the age of the tree pieces can be calculated.

EXAMPLE 9.14

Keezhadi (கீழடி), a small hamlet, has become one of the very important archeological places of Tamilandu. It is located in Sivagangai district. A lot of artefacts (gold coins, pottery, beads, iron tools, jewellery and charcoal, etc.) have been unearthed in Keezhadi which have given substantial evidence that an ancient urban civilization had thrived on the banks of river Vaigai. To determine the age of those materials, the charcoal of 200 g sent for carbon dating is given in the following figure (b). The activity of ${}^{14}_{6}C$ is found to be 37 decays/s. Calculate the age of charcoal.



Figure (a) Keezhadi – excavation site



Figure (b) – Characol which was sent for carbon dating

Solution

To calculate the age, we need to know the initial activity (R_0) of the charcol (when the sample was alive).

The activity R of the sample

$$R = R_0 e^{-\lambda t} \qquad (1)$$

To find the time t, rewriting the above equation (1), $e^{\lambda t} = \frac{R_0}{R}$

By taking the logarithm on both sides, we get $t = \frac{1}{\lambda}\ln\left(\frac{R_0}{R}\right)$ (2)

Here R = 38 decays/s=38 Bq.

To find decay constant, we use the equation

$$\lambda = \frac{0.6931}{T_{1/2}} = \frac{0.6931}{5730\,\text{yr} \times 3.156 \times 10^7\ \text{s/yr}}$$

$[\because\ 1\text{yr} = 365.25 \times 24 \times 60 \times 60\ \text{s} = 3.156 \times 10^7\ \text{s}]$

$$\lambda = 3.83 \times 10^{-12}\, s^{-1}$$

To find the initial activity R_0, we use the equation $R_0 = \lambda N_0$. Here N_0 is the number of carbon-14 atoms present in the sample when it was alive. The mass of the charcol is 200 g. In 12 g of carbon, there are 6.02×10^{23} carbon atoms. So 200g contains

$$\frac{6.02 \times 10^{23}\ \text{atoms}/mol}{12g/mol} \times 200 \approx 1 \times 10^{25}\ \text{atoms}$$

When the tree(sample) was alive, the ratio of ${}^{14}_{6}C$ to ${}^{12}_{6}C$ is 1.3×10^{-12}. So the total number of carbon-14 atoms is given by

$$N_0 = 1 \times 10^{25} \times 1.3 \times 10^{-12} = 1.3 \times 10^{13}\ \text{atoms}$$

The initial activity

$$R_0 = 3.83 \times 10^{-12} \times 1.3 \times 10^{13} \approx 50\,\text{decays}/s$$
$$= 50\,Bq$$

By substituting the value of R_0 and λ in the equation (2), we get

XII_Physics_Vol-2_UNIT-9_EM.indd 178 19-12-2022 16:06:44